SOASMI KOHLI

Software Developer | AI/ML Engineer
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PROFESSIONAL SUMMARY

Final, year B.Tech (AI & ML) undergraduate with hands, on software engineering experience across three paid/credited internships — building production Flutter applications, Python/FastAPI REST backends, containerised services, and real, time data pipelines. Demonstrated ability to deliver end, to, end, maintainable software using OOP, Git workflows, and Agile practices. AI/ML background (DRDO, IEEE publication) is an engineering differentiator, not the primary identity. Seeking to bring full, stack and backend development skills to Penthara Technologies.

EDUCATION

B.Tech — CSE (Artificial Intelligence & Machine Learning) *2023 – 2027*
Vivekananda Institute of Professional Studies, GGSIPU | CGPA: 9.77

TECHNICAL SKILLS

Languages & Web: JavaScript, HTML/DOM, CSS | Python, Dart, Java, SQL, C
Frameworks & APIs: REST APIs, FastAPI, Flutter, Node.js, Riverpod
Dev Tools: Git & GitHub, Docker, NPM, VS Code, Android Studio, Agile / Sprint, based delivery
Databases: MySQL, SQLite, PostgreSQL, MongoDB, SQLAlchemy / Alembic
Software Engineering: OOP, Clean Architecture, DSA, Cross, Platform Development, Performance Optimisation
AI / ML (supporting): NLP, CNNs, Computer Vision, OpenCV, XGBoost, FAISS, SentenceTransformers, LlamaIndex

PROFESSIONAL EXPERIENCE

Cyber Security & Software Engineering Intern | CyberPeace *Apr 2025 – Jul 2025*

- Engineered and shipped a production, ready Flutter application deployed across web and mobile, achieving ~40% faster development velocity through reusable component architecture and clean separation of concerns.
- Built a modular Python/FastAPI REST backend for a phishing, detection pipeline (99%+ precision, ~45% fewer false positives), exposing clean API endpoints consumed by the Flutter frontend.
- Architected a Docker, based browser sandbox with strict session isolation to enforce security boundaries and operational robustness.

AI, ML & Software Engineering Intern (Paid) | DRDO *Jan 2026 – Apr 2026*

- Designed and integrated a production, grade sensor data pipeline for geomagnetic anomaly analysis: built modular data, ingestion, processing, and validation layers; improved simulated detection accuracy by ~80% through systematic testing and architectural refinement.
- Developed structured backend data workflows for quantum magnetometer, based sensing, producing reliable, queryable outputs for GNSS, denied underwater detection use cases.
- Applied OOP, driven architecture for modularity, testability, and fault tolerance across CNN and hybrid ML model integrations on noisy real, world sensor inputs.

Product Management Intern | Foruppo *Jun 2024 – Oct 2024*

- Contributed to live Agile product cycles — sprint planning, feature scoping, and iterative delivery — collaborating cross, functionally with engineering and product teams.

PROJECTS

CommUnity — WhatsApp Relationship Management System | *Flutter · Node.js · REST APIs · Supabase · PostgreSQL · Git*

- Built a full, stack web application: Flutter Web frontend (Riverpod, GoRouter) + Node.js backend integrating the Baileys WhatsApp API for automated and manual messaging workflows.
- Implemented Supabase/PostgreSQL backend with role, based authentication, real, time event handling, and a modular service architecture separating messaging, auth, and data layers; version, controlled via Git.

EchoSense — Multimodal Assistive Platform | *Python · NLP · Speech Processing · Gesture Recognition | IEEE Hackathon Winner 2025*

- Architected a real, time multimodal communication platform integrating NLP, gesture recognition, and speech, processing modules into a unified, extensible software pipeline.
- Won IEEE Hackathon 2025 (multi, university competition); published in IEEE Conference Proceedings, CISES 2025.

Document Intelligence Platform | *Python · FastAPI · FAISS · SentenceTransformers · SQLite · LlamaIndex*

- Architecting a fully offline Flutter Desktop + FastAPI application that ingests PDF/DOCX files, chunks and indexes embeddings via FAISS CPU, and drives LLM, powered report generation with no cloud dependency.

Database Management & Dataset Handling System | *Flutter · SQLite · Python*

- Developed a full, stack Flutter + SQLite application serving 12,500+ users: efficient CRUD, optimised structured queries, and a UI layer fully decoupled from data logic for maintainable, extensible architecture.

ACHIEVEMENTS & PUBLICATIONS

- Winner — IEEE Hackathon 2025 (EchoSense) | IEEE CISES 2025 Publication — EchoSense | Book Chapter: LLMs, LVMs & Big Data in Healthcare
- Harvard Aspire Leaders Program (ALP) Scholar 2025 | SheCodes Foundation Scholar 2025 | NPTEL STAR ×3